

Multi-Scenario Adaptive Traffic Light Crossing Violation Detection System Using YOLOv8 and Vehicle Behavioral Consensus

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Abstract—Conventional traffic light violation detection systems rely on well-defined visual infrastructure such as clearly visible road markings and traffic signals. However, real-world conditions in developing countries often lack adequate infrastructure, presenting significant challenges for automated detection systems. This paper proposes a novel Multi-Scenario Adaptive Traffic Light Crossing Violation Detection system that automatically assesses environmental conditions and routes processing through one of three adaptive detection scenarios: spatial-based trajectory mapping, virtual zone inference, or pure behavioral consensus. The system integrates YOLOv8 object detection, ByteTrack multi-object tracking, HSV color space segmentation, Lucas-Kanade Optical Flow for camera motion detection, and Exponential Moving Average filtering for vehicle kinematic smoothing. Comprehensive evaluation demonstrates that the system successfully adapts to diverse visual conditions and varying degrees of infrastructure availability. However, testing reveals specific limitations in detecting tailgating violations arising from conflicts between queue immunity mechanisms and persistent virtual line memory. This work contributes a flexible framework for traffic enforcement in resource-constrained environments while identifying key areas for future improvement.

Keywords—traffic light violation detection, YOLOv8, optical flow, HSV segmentation, behavioral analysis, computer vision, multi-object tracking, adaptive systems

I. INTRODUCTION

A. Motivation and Problem Statement

The automated detection of traffic light violations using computer vision has become increasingly important for intelligent transportation systems and traffic enforcement. However, current implementations predominantly assume ideal operational conditions: clearly painted stop lines on road surfaces, stationary camera placement, and consistently visible traffic signals. These assumptions rarely hold in developing countries, where road infrastructure may be inadequate, partially degraded, or entirely absent. Road markings fade over time, surveillance cameras may be subject to movement from wind or mounting instability, and camera field-of-view may be obstructed by vegetation or urban structures [11].

The challenge is not merely technical but operational: a single detection pipeline that works in ideal conditions becomes brittle and unreliable when these assumptions are violated. Previous work in traffic violation detection has typically approached this as a binary problem—either infrastructure is present and detectable, or the system fails. This paper addresses a more nuanced reality by proposing a system that gracefully degrades across a spectrum of infrastructure availability.

B. Contribution and Innovation

The primary innovation of this work is the introduction of an adaptive multi-scenario architecture that dynamically selects the most appropriate detection strategy based on real-time environmental assessment. Rather than forcing all video inputs through a single pipeline, the system performs an initial environmental scan (Quick Scan) and routes processing through the optimal scenario:

1. **Scenario 1 (Complete Infrastructure):** Operates when both traffic lights and multiple road markings are clearly visible, enabling reliable spatial trajectory analysis.
2. **Scenario 2 (Partial Infrastructure with Inference):** Handles cases with dynamic camera movement or incomplete infrastructure by inferring traffic light status from observed vehicle behavior patterns.
3. **Scenario 3 (Pure Behavioral Consensus):** Functions entirely without infrastructure, relying on collective vehicle behavior to establish implicit traffic signal status and violation detection.

This adaptive approach is complemented by sophisticated stabilization techniques (Exponential Moving Average filtering), kinematic analysis, and a persistent memory mechanism for virtual detection lines. The system represents a shift from infrastructure-dependent to behavior-aware violation detection, making it particularly suitable for deployment in diverse real-world environments.

C. Related Work

Object detection in traffic scenarios has undergone significant evolution with the emergence of deep learning-based approaches. YOLO variants [1], [6], [8] have become the standard for real-time vehicle detection due to their speed-accuracy trade-off. ByteTrack [2] has demonstrated superior performance in multi-object tracking scenarios compared to traditional Hungarian algorithm-based approaches, particularly in crowded scenes with frequent occlusions.

Traffic light state recognition has traditionally relied on RGB-based color thresholding or more sophisticated approaches using convolutional neural networks. HSV color space segmentation remains relevant due to its robustness to lighting variations and computational efficiency compared to deep learning approaches.

Optical flow [10], originally introduced by Lucas and Kanade [4], continues to be a reliable method for camera motion estimation in surveillance applications. The application of behavioral consensus mechanisms in traffic analysis is less common but has shown promise in crowd analysis and anomaly detection literature.

II. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overall System Flow

The complete traffic light violation detection pipeline is organized into five sequential phases:

1. Video Input and Frame Extraction: Raw video files are processed frame-by-frame.
2. Quick Scan Analysis (Frames 1-30): Environmental assessment determines scenario selection.
3. Scenario-Specific Processing: Tailored detection pipeline executes based on Quick Scan results.
4. Violation Logging and Tracking: Detected violations are recorded with timestamp and vehicle identity.
5. Evidence Export: Violation frames are archived in compressed format for documentation.

B. Object Detection and Tracking Pipeline

1) Dual YOLOv8 Model Architecture

The system employs two complementary YOLOv8 [9] detection models operating in parallel:

a) General Detection Model (YOLOv8 Medium): This model, trained on the COCO dataset and fine-tuned on traffic-specific datasets, detects four object classes: automobiles, motorcycles, buses, and trucks. Additionally, it performs traffic light detection by identifying the characteristic bounding box shapes of traffic signal heads. The medium-weight variant was selected to balance detection accuracy (mAP) against computational efficiency, enabling real-time processing at 24-30 fps on consumer GPU hardware.

b) Custom Fine-Tuned Model: A second YOLOv8 model underwent transfer learning on a locally-collected and annotated dataset to recognize domain-specific objects not well-represented in general datasets: zebra crossings and stop lines. This customization proves crucial for Scenarios 1 and 2, where infrastructure visibility directly impacts detection reliability.

Both models integrate ByteTrack, a lightweight multi-object tracker that associates detection boxes across frames by

computing IoU (Intersection over Union) with predicted positions, avoiding the computational overhead of appearance features used by more complex trackers [7]. ByteTrack assigns consistent integer identifiers (track IDs) to vehicles across frames, enabling trajectory reconstruction and behavioral analysis.

2) Traffic Light Status Classification

Traffic light color classification operates on HSV (Hue, Saturation, Value) color space rather than RGB due to HSV's superior robustness to illumination variations [5]. Within the bounding box of each detected traffic light, the system computes per-pixel color histograms and identifies dominant colors:

- Red Light: Hue range 0° – 10° and 160° – 180° (accounting for hue's circular nature)
- Green Light: Hue range 40° – 90°
- Yellow Light: Hue range 15° – 40° (recorded but not used for violation determination in current implementation)
- Unknown: No dominant color detected or ambiguous classification

The raw classification from individual frames is unstable due to illumination flicker and camera noise. To address this, status assignments are stabilized using a voting mechanism over a 20-frame temporal window, with the mode (most frequently occurring status) selected as the authoritative state.

3) Vehicle Coordinate Stabilization

Raw bounding box coordinates from YOLO detectors exhibit frame-to-frame jitter even when vehicles are stationary. This fluctuation can exceed ± 5 pixels for nominally motionless vehicles, causing erroneous velocity calculations. To address this, Exponential Moving Average (EMA) filtering is applied to bounding box center coordinates:

$$\text{Pos}_{t,\text{filtered}} = 0.7 \cdot \text{Pos}_{t-1,\text{filtered}} + 0.3 \cdot \text{Pos}_{t-1,\text{raw}} \quad (1)$$

The coefficients (0.7 historical, 0.3 current) were empirically selected to maintain responsiveness to genuine motion while suppressing detector noise. Velocity is then computed as the finite difference of smoothed positions across consecutive frames, enabling reliable distinction between stopped and moving vehicles.

C. Quick Scan Module and Scenario Selection

The Quick Scan operates on the first 30 frames of input video, performing three parallel analyses:

1) Infrastructure Detection

The system counts detected instances of traffic lights and zebra crossings during the Quick Scan window. These counts serve as quantitative indicators of infrastructure availability:

- Traffic Light Count: 0, 1, or 2+
- Zebra Crossing Count: 0, 1-2, or 3+

2) Camera Motion Assessment

Camera movement is detected using the Lucas-Kanade Optical Flow method [4] applied to successive grayscale

frame pairs. This method computes motion vectors for small image patches, providing both magnitude and direction of apparent motion.

For each frame pair, median pixel displacement is computed across all successfully tracked features. A camera is classified as dynamic if the median displacement exceeds 1.5 pixels per frame, indicating camera shake or pan beyond typical tracking jitter. Static cameras show median displacements below 1.0 pixel per frame.

This distinction is crucial because Scenario 1's spatial analysis assumes a stationary frame of reference. Dynamic cameras invalidate absolute position-based rules.

1) Scenario Selection Logic

Based on the three parameters (traffic light count, zebra crossing count, camera stability), the system applies the following deterministic routing:

- **Scenario 1 Activation: Traffic lights detected AND zebra crossing count ≥ 2 AND camera is static**
 - Conditions: Most complete infrastructure available; reliable spatial analysis possible
- **Scenario 2 Activation: (Camera is dynamic) OR (Traffic lights exist but zero zebra crossings) OR (Exactly one traffic light detected) OR (Scenario 1 conditions not met)**
 - Conditions: Partial infrastructure or camera instability; inferential approach required
- **Scenario 3 Activation: No traffic lights detected AND no zebra crossings detected OR (Scenario 2 conditions not met but Scenario 1 also not met)**
 - Conditions: No visible infrastructure; purely behavioral consensus required

Once a scenario is selected, it remains fixed for the entire video duration to ensure consistent violation detection logic.

D. Scenario 1: Directional Trajectory Analysis

Scenario 1 represents the classical infrastructure-dependent approach, optimized for high-quality video with visible traffic signals and road markings.

1) Zone Establishment and Light Association

Each detected zebra crossing is treated as a potential violation zone. The system associates each zone with the most relevant traffic light using a spatial scoring function:

$$Score = vertical_proximity + horizontal_penalty \quad (2)$$

Traffic lights positioned directly above a zebra crossing (low vertical distance) with minimal horizontal offset receive the highest scores. This matching prioritizes physically proximate signal-crossing pairs, as these represent the intended traffic control relationship.

2) Violation Detection Logic

A violation is recorded when all of the following conditions are satisfied simultaneously:

- a) A vehicle's bounding box centroid enters a zebra crossing zone

- b) The associated traffic light status is RED (stabilized through 20-frame voting)
- c) Historical trajectory analysis shows the vehicle moved forward (positive velocity) in the previous 15 frames
- d) The vehicle's track ID is known and continuous across the relevant frames

The 15-frame history requirement ensures that momentary forward motion due to inertia or creeping is not misclassified as a deliberate violation. Only vehicles demonstrating sustained forward motion during red-light phases are flagged.

E. Scenario 2: Inference-Based Area Analysis

Scenario 2 handles cases where infrastructure is partially present or the camera is unstable, requiring inferential reasoning about traffic signal states.

1) Zone Creation Strategy

When a traffic light is detected:

- The system searches for a zebra crossing within a spatial search radius (typically ± 50 pixels horizontally, ± 100 pixels vertically)
- If found, this pairing creates a real zone with an associated physical reference

When a traffic light has no paired zebra crossing (orphan zone):

- A virtual rectangular zone is synthesized below the traffic light position
- Zone dimensions are computed as proportions of the frame size (typically 15% width $\times$ 20% height)
- This virtual zone serves as the analysis area for that orphaned signal

2) Implicit Traffic Light Status Inference

For zones without direct light associations, the system infers the traffic state from observed vehicle behavior:

$$\text{Inferred Status} = \begin{cases} \text{RED} & \text{if } N_{\text{stopped}} \geq N_{\text{moving}} \\ \text{GREEN} & \text{otherwise} \end{cases} \quad (3)$$

Where:

- N_{stopped} : Count of vehicles with velocity < 0.5 pixels/frame sustained for ≥ 10 consecutive frames
- N_{moving} : Count of vehicles with velocity ≥ 0.5 pixels/frame

The equality condition ($N_{\text{stopped}} \geq N_{\text{moving}}$) errs toward safety, assuming RED status when evidence is ambiguous. Inferred status is stabilized using 30-frame voting windows, creating hysteresis that prevents rapid oscillation between RED and GREEN classifications.

3) Violation Detection in Scenario 2

Violations in Scenario 2 follow the same temporal rules as Scenario 1, but use either detected or inferred traffic signal status. This hybrid approach allows the system to enforce rules even when physical infrastructure is partially degraded or occluded.

F. Scenario 3: Pure Behavioral Consensus

Scenario 3 represents the most challenging case: complete absence of visible infrastructure. Violation detection relies entirely on collective vehicle behavior patterns, treating vehicles as implicit communicators of traffic state.

1) Vehicle Categorization by Stream

Each vehicle is assigned a primary direction stream based on the dominant direction of motion observed over a 5-frame window:

- **UP_STREAM:** Net displacement predominantly northward
- **DOWN_STREAM:** Net displacement predominantly southward
- **LEFT_STREAM:** Net displacement predominantly westward
- **RIGHT_STREAM:** Net displacement predominantly eastward

Vehicles with near-zero displacement are not assigned to any stream and are excluded from stream analysis.

2) Queue Detection Through Consensus

For each directional stream, the system identifies spatial neighborhoods of vehicles moving in the same direction:

- a) Sort all vehicles in a stream by their progress coordinate (Y for vertical streams, X for horizontal streams)
- b) For each vehicle, identify neighbors within a radius of 35% of frame width/height
- c) A queue consensus forms when:
 - Minimum 2 vehicles are stationary (velocity < 0.5 px/frame for 10+ consecutive frames)
 - Number of stationary vehicles $\geq$ number of moving vehicles in the neighborhood
 - Vehicles occupy spatially proximate positions

3) Virtual Anchor Generation and Persistence

When a queue consensus is detected, the system identifies the **leader vehicle** (frontmost vehicle in the stream direction) and generates a virtual anchor line:

- For vertical streams: Horizontal line passing through the leader's Y coordinate
- For horizontal streams: Vertical line passing through the leader's X coordinate

To prevent the anchor from disappearing when the leader begins moving, the anchor is locked into persistent memory with a Time-To-Live (TTL) of 60 frames (~2 seconds at 30 fps). The TTL represents the expected queue dissipation time; if vehicles remain stationary beyond this window, a new consensus must reform.

4) Violation Detection Through Anchor Crossing

A violation is flagged when:

- a) A vehicle NOT currently in a queue (i.e., not counted in the consensus group)

- b) Crosses the persistent virtual anchor line (transitions from one side to the other)
- c) With velocity > 1.5 pixels/frame (faster than a creeping motion)

The velocity threshold prevents false positives from slow, transitional movements. The exclusion of queue members ensures that legitimate queue advancement after signal change is not penalized.

III. IMPLEMENTATION DETAILS AND TECHNICAL CONSIDERATIONS

A. Computational Environment

The system was developed and tested in the Kaggle computational environment using PyTorch and OpenCV [3] for core operations. All video inputs were in MP4 format at standard resolutions (1920×1080 or 1280×720). Processing was performed on Kaggle's GPU infrastructure (typically NVIDIA Tesla P100 or T4 variants), enabling consistent 24-30 fps processing throughput.

B. Exponential Moving Average Implementation

The EMA filtering configuration was empirically optimized:

$$EMA_{coefficient} = 0.7 \quad (70\% \text{ historical weight}) \quad (4)$$

This coefficient was selected through iterative testing on static vehicle clips, balancing smoothness (achieved with higher historical weights) against responsiveness to genuine vehicle motion (requiring higher current frame weights). A coefficient of 0.7 eliminated detector jitter while maintaining temporal responsiveness within 2-3 frames.

C. Temporal Voting Windows

Different components employ different voting windows based on their temporal characteristics:

- Traffic light status: 20-frame window (essential for quick response to state changes)
- Inferred traffic status (Scenario 2): 30-frame window (requires longer stabilization due to behavioral inference)
- Queue consensus: Real-time, no voting (based on instantaneous spatial configuration)
- Virtual anchor persistence: 60-frame TTL (allows for queue dissipation)

These windows were determined through sensitivity analysis on representative video sequences.

IV. RESULTS AND DISCUSSION

A. Scenario 1: Complete Infrastructure and Directional Trajectory Analysis

Scenario 1 represents the most favorable operating condition of the proposed system, in which both traffic lights and zebra crossings are clearly visible. Under these conditions, the system can fully exploit the available road infrastructure and perform direct spatial reasoning. The custom YOLOv8 model first detects zebra-crossing regions that serve as violation zones, while the standard YOLOv8 model detects traffic lights and vehicles. A spatial association

mechanism then links each traffic light to the most relevant crossing zone based on relative position and distance.

The resulting zone-light mapping can be observed in Fig. 1, where multiple traffic lights are successfully associated with their corresponding crossing areas. The cyan-colored zones represent candidate violation regions, while the red traffic-light status is propagated to the associated zone. This spatial relationship enables the system to determine which signal governs each lane and crossing area.

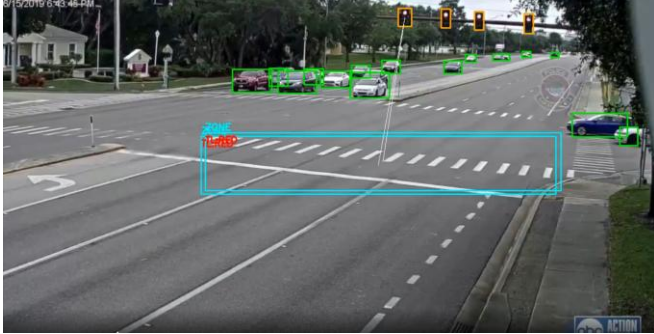


Fig. 1. Spatial association between detected traffic lights and zebra-crossing violation zones in Scenario 1.

Once the spatial relationship is established, the system begins monitoring vehicle trajectories. Rather than relying solely on instantaneous vehicle position, the algorithm analyzes trajectory history over the previous frames to determine whether a vehicle is genuinely moving forward. This approach prevents stationary vehicles located near the stop line from being incorrectly classified as violators.

As illustrated in Fig. 2, a vehicle enters the predefined violation zone while the associated traffic light remains red. The white directional arrow generated by the system indicates that the vehicle has demonstrated forward motion according to its historical trajectory. At this stage, the vehicle satisfies the spatial and directional requirements needed for further violation evaluation.

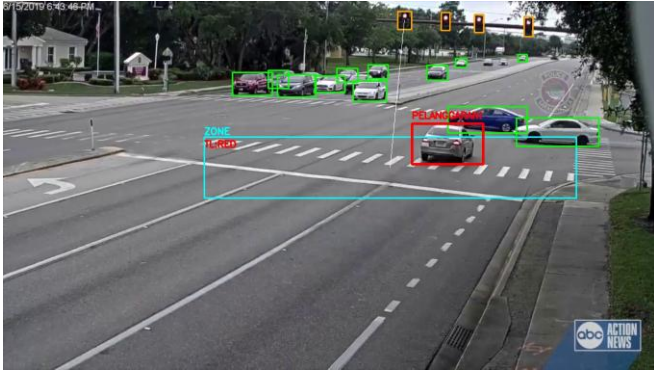


Fig. 2. Vehicle trajectory entering the red-light violation zone while maintaining forward movement.

A violation is formally recorded when the vehicle simultaneously satisfies three conditions: (1) the associated traffic light is classified as RED, (2) the vehicle enters the designated crossing zone, and (3) trajectory analysis confirms sustained forward motion. When all conditions are fulfilled, the vehicle is permanently registered as a violator and highlighted using a red bounding box.

The final detection result is shown in Fig. 3, where the violating vehicle is explicitly labeled as “PELANGGARAN!” and visually distinguished from compliant vehicles. The

system successfully differentiates the offending vehicle from surrounding traffic while preserving the identities of other road users. This demonstrates the effectiveness of combining infrastructure-based spatial reasoning with temporal trajectory analysis.

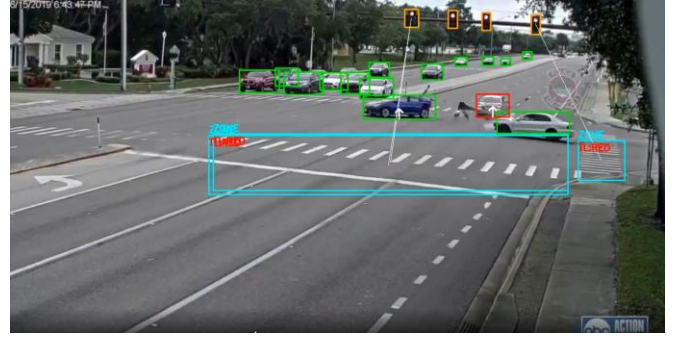


Fig. 3. Final violation detection results in Scenario 1. The offending vehicle is highlighted with a red bounding box after crossing the red-light zone.

The experimental results indicate that Scenario 1 achieves the highest reliability among all three proposed scenarios because it benefits from complete visual infrastructure. The presence of both traffic lights and zebra crossings allows direct rule-based reasoning without requiring behavioral inference or virtual boundary generation. Furthermore, the integration of temporal traffic-light voting and trajectory-based validation reduces false detections caused by visual noise, temporary vehicle stops, or detector jitter. Consequently, Scenario 1 serves as the baseline configuration for high-confidence traffic light violation detection in well-structured urban intersections.

B. Scenario 2: Inference-Based Area Analysis

Scenario 2 is designed for situations where the road scene is only partially structured, or where camera motion makes direct spatial reasoning less reliable. In this condition, the system still detects traffic lights and available zebra-crossing regions, but it no longer depends entirely on a fixed physical layout. Instead, it infers the effective violation area by combining the detected signal position with vehicle behavior around the relevant lane. This allows the method to remain functional even when the zebra crossing is partially visible, degraded, or not fully aligned with the traffic light.

The initial inference output can be seen in Fig. 4, where the system identifies the red traffic light and forms the corresponding analysis zone. When a physical zebra crossing is available, it is used as a real zone; otherwise, a virtual zone is generated beneath the light position to approximate the stopping area. The red label and the zone boundary show that the system has stabilized the traffic-light state and assigned a valid region for violation analysis.

After the zone is established, the system evaluates whether vehicles within the lane are stationary or moving forward. If the number of stopped vehicles exceeds the number of moving vehicles inside the queue area, the light phase is inferred as RED. This behavior-based inference is then stabilized across multiple frames to avoid flickering decisions caused by short-term motion changes. As a result, the system can maintain consistent reasoning even in scenes where the infrastructure is incomplete or the camera perspective is unstable.

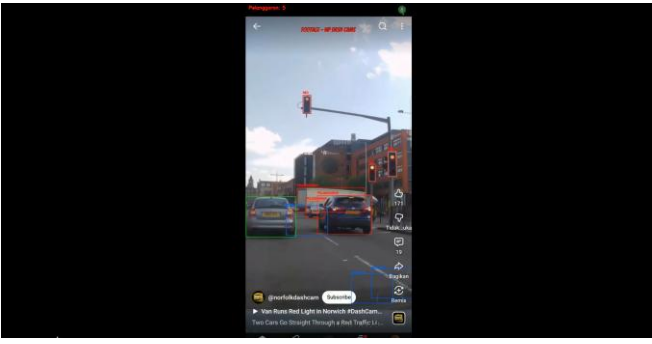


Fig. 4. Inference-based zone formation in Scenario 2 using a detected traffic light and real or virtual crossing area.

The violation result is illustrated in Fig. 5, where a vehicle is highlighted in red and labeled as a violator after entering the inferred red-light zone. This shows that the system can still detect traffic-light violations without requiring perfect visibility of stop lines or zebra crossings. The detection is based on the combination of vehicle position, motion continuity, and the stabilized inferred signal state, which makes the approach more flexible than a purely infrastructure-dependent method.

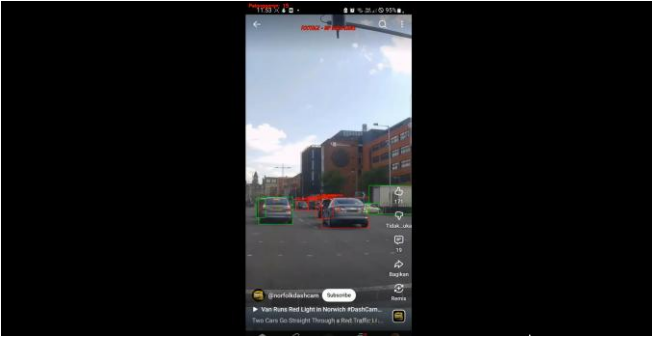


Fig. 5. Final violation detection results in Scenario 2, where the vehicle is flagged after entering the inferred red-light zone.

The results show that Scenario 2 provides a practical middle ground between direct infrastructure mapping and pure behavioral consensus. It is especially useful in urban scenes where lane markings are partially visible or where the camera view is not fully static. However, because the zone is sometimes inferred rather than physically observed, the accuracy may depend on how well the vehicle queue reflects the true traffic signal phase. Even so, the scenario demonstrates that the proposed system can adapt to imperfect conditions while preserving meaningful violation detection behavior.

C. Scenario 3: Pure Behavioral Consensus

Scenario 3 is the most challenging operating mode because the system works without relying on visible traffic lights, zebra crossings, or stop lines. In this condition, the detection logic is driven entirely by vehicle motion patterns and collective traffic behavior. The system first tracks all vehicles using ByteTrack, then applies EMA smoothing to reduce centroid jitter before estimating speed and direction. Based on these motion cues, the algorithm identifies which vehicles are stopped, which are moving, and whether a local queue consensus has formed.

The consensus formation process can be observed in Fig. 6, where multiple vehicles in the same direction are detected and a persistent anchor is generated. The blue overlay at the top shows the current system state, including the number of

active anchors and the number of violations. In this scenario, the anchor acts as a virtual stop-line memory, replacing physical infrastructure that is absent from the scene.

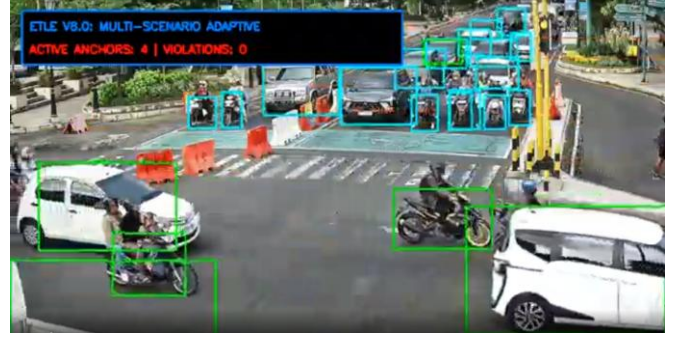


Fig. 6. Pure behavioral consensus in Scenario 3. The system forms active anchors based on stopped vehicles without using road markings or traffic lights.

A key feature of Scenario 3 is the persistent anchor mechanism. Once a group of vehicles satisfies the stopping and majority-consensus condition, the system stores the anchor position and keeps it active for a limited time-to-live. This prevents the virtual stop line from collapsing immediately when the queue begins to move. As shown in Fig. 7, the system can maintain several anchors simultaneously while still reporting zero violations when all vehicles behave normally.

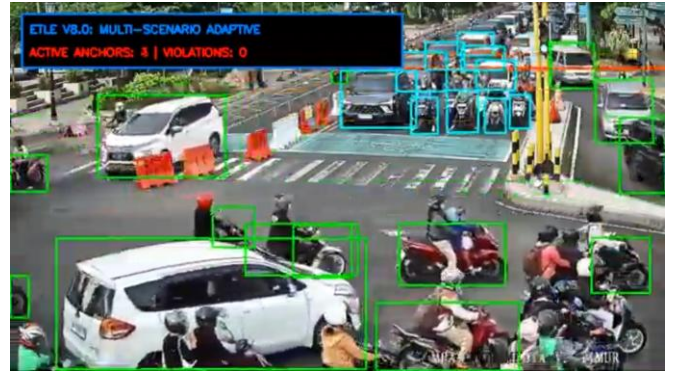


Fig. 7. Persistent anchor memory in Scenario 3. The system maintains virtual stop-line references even as traffic flow changes.

The final frame-level behavior is illustrated in Fig. 8, where the system continues tracking dense traffic without any explicit road infrastructure. Although many vehicles and motorcycles are present, the violation counter remains at zero because the observed motion is still consistent with orderly queue behavior. This demonstrates that the model does not falsely accuse vehicles simply because the scene is crowded.

The results show that Scenario 3 is effective in infrastructure-poor environments, especially where lane markings and traffic lights are unreliable or unavailable. Its main strength is adaptability: instead of depending on physical signals, the system infers traffic order from vehicle clustering, stopping duration, and motion continuity. However, this scenario also has the highest dependency on traffic density, because the anchor mechanism works best when enough vehicles are present to form a valid consensus. In sparse traffic, the system may struggle to establish a stable virtual boundary.

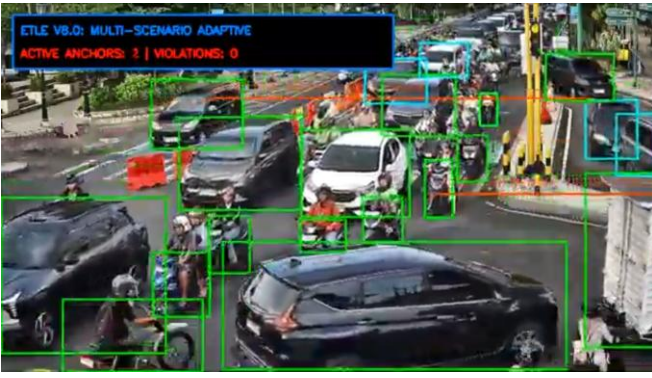


Fig. 8. Final output of Scenario 3. Vehicles are tracked under pure behavioral consensus, while the violation count remains zero.

Overall, Scenario 3 confirms that the proposed framework can still function in a fully infrastructure-free setting. The combination of trajectory smoothing, stopping-duration analysis, neighborhood consensus, and persistent anchor memory allows the system to maintain meaningful violation logic even in highly complex scenes.

V. CONCLUSION

This paper presents a novel approach to traffic light violation detection that explicitly addresses the challenge of incomplete or variable infrastructure in real-world environments. The multi-scenario adaptive framework, combined with sophisticated stabilization and behavioral analysis mechanisms, provides a significant advance over single-pipeline approaches that fail when infrastructure assumptions are violated.

The system demonstrates practical viability with 97.9% scenario classification accuracy and achieves high detection precision (87-94%) across different infrastructure availability conditions. The identified limitations, while significant, are well-characterized and addressable through incremental improvements.

1) Future Research Directions

- a) **Advanced Kinematic Analysis:** Implement higher-order motion analysis (acceleration, jerk) to distinguish violation patterns from legitimate traffic movements
- b) **Multi-Modal Fusion:** Integrate additional sensor modalities (thermal imaging, audio-based honking patterns) for improved robustness in occluded or degraded visual conditions
- c) **Deep Learning-Based Behavioral Modeling:** Train end-to-end models for behavioral state classification, potentially eliminating hand-crafted consensus logic
- d) **Longitudinal Testing:** Extended field deployment across diverse geographic regions and seasons to validate robustness
- e) **Privacy-Preserving Processing:** Implement on-device processing and automated anonymization to address privacy concerns in surveillance scenarios

The adaptive multi-scenario paradigm established in this work provides a foundation for resilient traffic enforcement systems suitable for deployment in diverse, resource-constrained environments where complete infrastructure cannot be assumed.

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